

ShopZon E-Commerce Portal

Comprehensive Full-Stack Technical Project Report & Architecture Overview

1. Project Overview

ShopZon is a fully functional, responsive, full-stack e-commerce web application engineered to handle end-to-end online retail operations. The system incorporates a robust public storefront designed for seamless product browsing, dynamic filtering, cart management, and secure checkout, alongside an advanced administrative dashboard for overseeing inventory metrics, tracking live revenue, and managing global catalog listings.

2. Technology Stack & Architecture

The application implements a modern decoupled architecture, separating the high-performance client interface from scalable backend services and cloud database infrastructure:

Frontend Client:	React powers the dynamic component-based single-page user interface, styled utilizing Tailwind CSS to ensure responsive rendering across mobile, tablet, and desktop viewports.
Backend API:	Built with Python & Flask , providing RESTful API endpoints, secure authentication routing, and CORS handling via Flask-CORS .
Cloud Database:	Hosted on Aiven Managed MySQL , offering reliable cloud database persistence, high availability, and secure connection pooling via PyMySQL .
Cloud Deployment:	Frontend deployed on Vercel for global edge delivery; backend deployed on Render ; version control managed via GitHub .

3. Core Features & System Capabilities

- **Secure Authentication & Authorization:** Implements robust token/session validation workflows, preflight options support, and protected administrative routes.

- **Dynamic Product Catalog:** Advanced filtering capabilities across departments (such as Men's Clothing, Electronics, Jewelry, etc.) with real-time text search and instant inventory lookup.
- **Administrative Intelligence Dashboard:** Centralized control center monitoring key performance metrics including total revenue earned, active catalog classifications, sales volume, and dynamic product registration.
- **Cross-Environment Parity:** Fully configured for local development (via XAMPP/MySQL) and seamless production scaling on cloud infrastructure.

4. System Deployment & Links

The production-ready application is accessible globally via the official deployment links:

Live Web App: <https://shopzon-ecommerce.vercel.app>

Repository: [GitHub](#) / [shopzon-ecommerce](#)